

Student ID:		Lab Section:	
Name:		Lab Group:	

Experiment No. 1

Introduction to Series and Parallel Circuit Connections

Objective

This experiment aims to acquaint students with series and parallel circuit connections and to properly identify them on a breadboard or from a schematic diagram.

Theory

An electrical circuit is a continuous path through which electrical current flows. Amongst various circuit combinations, two prominent ones are called “Series” and “Parallel”. For a connection to be called “**Series**”, it must fulfill the following criteria:

- All the components must be connected *one after the other*.
- The *same current* must flow through all the components.

For instance, we have N resistors in the following circuit: $R_1, R_2, R_3, \dots, R_N$ connected one after another, and the same current I is flowing through them. All of these series resistors can be combined into just one equivalent resistance,

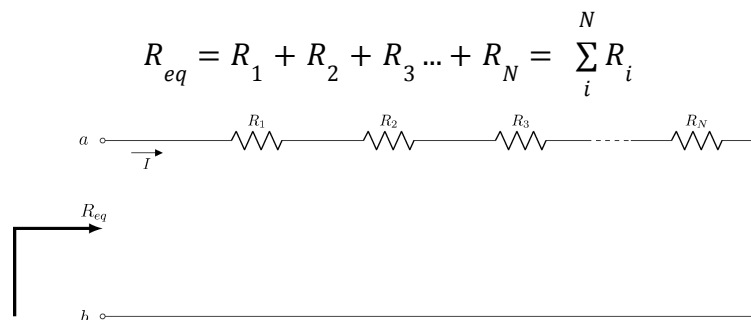


Figure: A series connection

Similarly, in a “**Parallel**” connection,

- All the components must be connected between the *same two nodes*
- The *same potential (voltage) drop* should exist across all the components.

For example, in the following figure, we have N resistors with resistances: $R_1, R_2, R_3, \dots, R_N$ connected at the same two nodes a and b . And therefore, the voltage drop across all the resistors is, $\Delta V = V_a - V_b$. Hence, we conclude that the resistors are connected in parallel. The equivalent resistance of these resistors is R_{eq} , where

$$\frac{1}{R_{eq}} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} \dots + \frac{1}{R_N} = \sum_i^N \frac{1}{R_i}$$

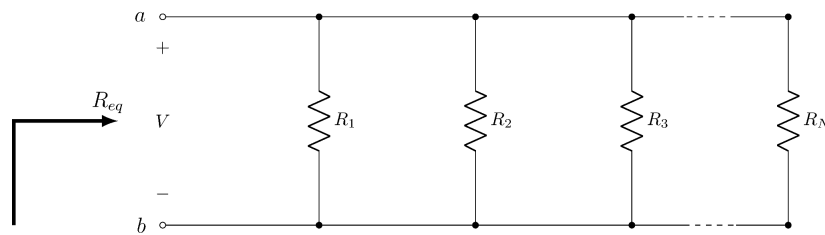


Figure: A parallel connection

In this experiment, we will learn how to connect circuits on breadboards and how to identify series and parallel connections.

Apparatus

- Multimeter
- Resistors
- DC power supply
- Breadboard
- Jumper wires

Procedures

- Measure the resistances of the provided resistors and fill in the data table.
- Construct **circuits 1, 2, and 3** on a breadboard. Try to use as few jumper wires as possible.
- Measure the equivalent resistance using a multimeter. **To do this, disconnect the power supply (if any) and connect the multimeter across the open terminals.**
- Apply a **6 V** potential drop across the terminals *a* and *b*. Use the DC power supply to connect the positive terminal to the node *a* and the negative terminal to the terminal.
- Measure the voltage and current across each resistor. Use a multimeter to measure the voltage and use Ohm's law to calculate the current through each resistor. Fill in the data tables.

Data Tables

Signature of Lab Faculty:

Date:

**** For all the data tables, take data up to three decimal places, round to two, then enter into the table.**

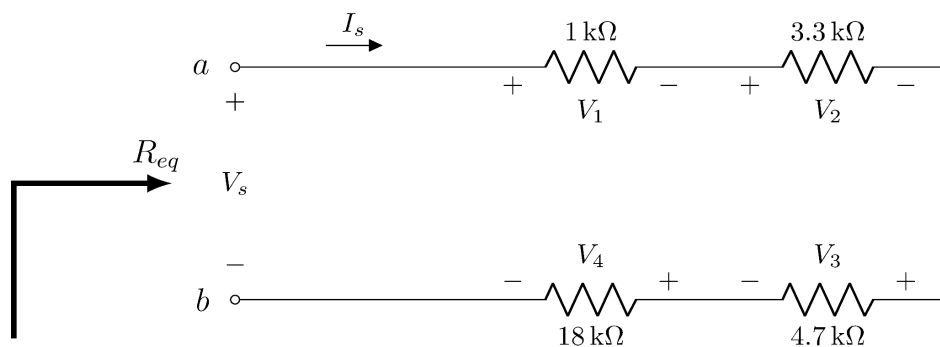
Table 0: Resistance Data

For all your future calculations, please use the observed values only (even for theoretical calculations).

Parameter	Expected Resistance	Observed Resistance (k Ω)	Parameter	Expected Resistance	Observed Resistance (k Ω)
R_1	1 k Ω		R_4	4.7 k Ω	
R_2	2.2 k Ω		R_5	18 k Ω	
R_3	3.3 k Ω				

Table 1: Data from Circuit 1

In the following table, V_1 is the voltage drop across the resistor R_1 , and I_1 is the current through it. A similar syntax applies to the remaining resistors. **For theoretical calculations, please note that, in a series connection, the supplied voltage will be divided proportionally to the resistances.** The voltage supplied to the complete circuit is denoted by V_s , and the current being supplied to the whole network is denoted as I_s . Also, calculate the percentage of error between experimental and theoretical values of R_{eq} .



Circuit 1

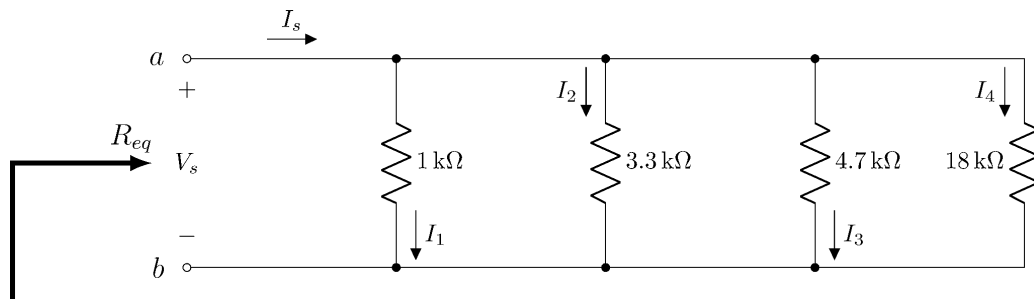
Observation	R_{eq} (k Ω)	V_s (V) (from dc power supply)	V_s (V) (using multimeter)	$I_s = \frac{V_s}{R_{eq}}$ (mA)	V_1 (V)	$I_1 = \frac{V_1}{R_1}$ (mA)	V_2 (V)	$I_2 = \frac{V_2}{R_2}$ (mA)	V_3 (V)	$I_3 = \frac{V_3}{R_3}$ (mA)	V_4 (V)	$I_4 = \frac{V_4}{R_4}$ (mA)
Experimental												
Theoretical												

$$\text{Percentage of error} = \left| \frac{\text{Experimental} - \text{Theoretical}}{\text{Theoretical}} \right| \times 100\%$$

Percentage of error in R_{eq} calculation = %

Table 2: Data from Circuit 2

In a parallel connection, all the voltage drops are the same across the components. Hence, we only need the supply voltage V_s . **However, the current across each component is inversely proportional to the resistance values.**



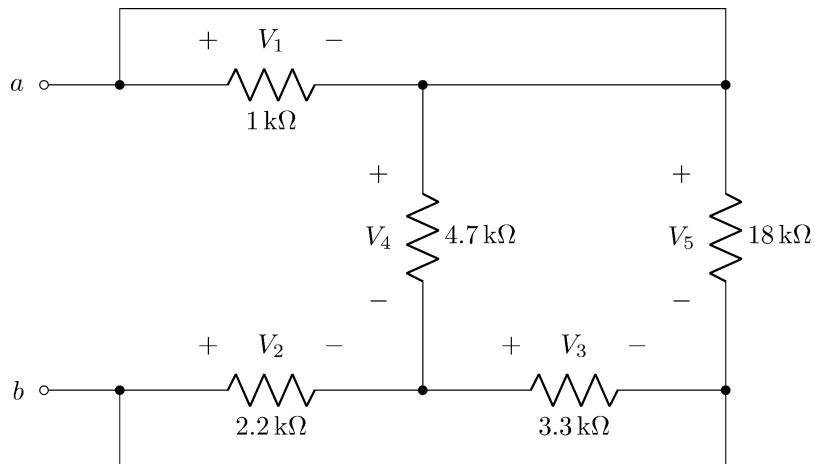
Circuit 2

Observation	R_{eq} (k Ω)	V_s (V) (from dc power supply)	V_s (V) (using multimeter)	$I_s = \frac{V_s}{R_{eq}}$ (mA)	$I_1 = \frac{V_s}{R_1}$ (mA)	$I_2 = \frac{V_s}{R_2}$ (mA)	$I_3 = \frac{V_s}{R_3}$ (mA)	$I_4 = \frac{V_s}{R_4}$ (mA)
Experimental								
Theoretical								

Percentage of error in R_{eq} calculation = %

Table 3: Data from Circuit 3

Collect the following data.



Circuit 3

Obs.	R_{ab} (kΩ)	V_s (V) (from dc power supply)	V_s (V) (using mult.)	$I_s = \frac{V_s}{R_{eq}}$ (mA)	V_1 (V)	$I_1 = \frac{V_1}{R_1}$ (mA)	V_2 (V)	$I_2 = \frac{V_2}{R_2}$ (mA)	V_3 (V)	$I_3 = \frac{V_3}{R_3}$ (mA)	V_4 (V)	$I_4 = \frac{V_4}{R_4}$ (mA)	V_5 (V)	$I_5 = \frac{V_5}{R_5}$ (mA)
Exp														
Theo														

Percentage of error in R_{eq} calculation = %

Based on the data taken for Circuit 3, what is your conclusion regarding the resistor connections? Identify which resistors are connected in series and which are connected in parallel.

Questions

1. Refer to the following illustration of the **Linear DC Power Supply** you used in the laboratory to answer the following questions—



- (a) Which of the operational modes of the power supply did you use for this experiment?
☐ Constant Voltage (C.V.) ☐ Constant Current (C.C.)
- (b) Maximum, how many voltages can we take from the power supply?
☐ 1 ☐ 2 ☐ 3 ☐ 4
- (c) If you are to take a voltage equal to 7 V, which of the channels can you use? Select all that apply—
☐ CH1 ☐ CH2 ☐ CH3 ☐ CH4
- (d) Which of the following is the voltage range for channel 3 (CH3)?
☐ 1 – 30 V
☐ 0 – 30 V
☐ 2.2 – 5.5 V
☐ 8 – 15 V
- (e) Which of the following is the voltage range for channel 4 (CH4)?
☐ 1 – 30 V
☐ 0 – 30 V
☐ 2.2 – 5.5 V
☐ 8 – 15 V

(f) What are the maximum current limits for CH1 and CH2, respectively?

- ☐ 1 A, 3 A
- ☐ 2 A, 3 A
- ☐ 3 A for both
- ☐ 1 A for both

(g) What are the maximum current limits for CH3 and CH4, respectively?

- ☐ 1 A, 3 A
- ☐ 2 A, 3 A
- ☐ 3 A for both
- ☐ 1 A for both

(h) Based on the functionality of the *channel selector buttons*, voltages for which pair of channels are we unable to display simultaneously? Select all that apply—

- ☐ CH1 & CH2 ☐ CH1 & CH3 ☐ CH1 & CH4
- ☐ CH2 & CH4 ☐ CH3 & CH4 ☐ CH2 & CH3

(i) If we want to set negative voltages to nodes with respect to a reference node (often called ground), can we do this using the power supply?

- ☐ Yes ☐ No

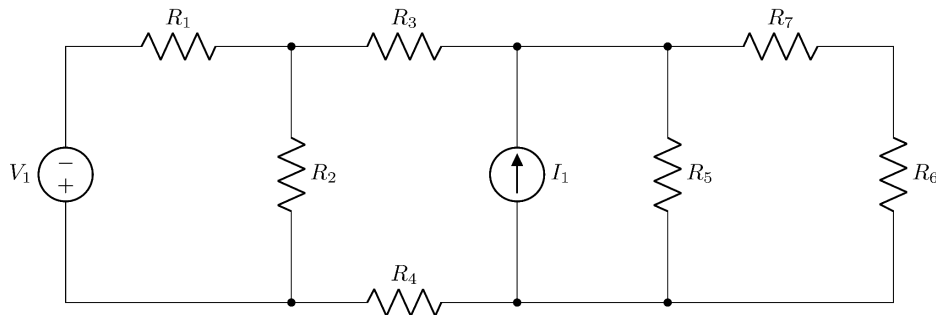
If yes, how?

(j) Check the squares adjacent to each of the following statements to indicate whether it is true or false:

- I. The bigger voltage and current knobs correspond to CH1 and CH2.
☐ True ☐ False
- II. The smaller voltage knobs are used to set voltages for CH3 and CH4.
☐ True ☐ False
- III. The current knobs are used to set the maximum limits of current.
☐ True ☐ False

- IV. The display shows the voltage-current for CH1 and CH2 only.
☐ True ☐ False
- V. Current values on the screen indicate the maximum current limit set in CH1 and CH2.
☐ True ☐ False
- VI. Only the maximum current limits for CH1 and CH2 are tunable.
☐ True ☐ False
- VII. The power supply doesn't provide any voltage for a channel if its current limit is set 0.
☐ True ☐ False
- VIII. Can this source be a constant current for applications such as recharging a battery?
☐ Yes ☐ No
- IX. Connecting CH1 and CH2 in series is a feature of the source.
☐ True ☐ False
- X. Pressing the "Output On" button makes the current values displayed equal to 0. It means the source is not supplying any current to the circuit connected to it.
☐ True ☐ False

2.



- (a) After taking voltage and current measurements in a laboratory for the circuit shown above, the currents through the R_4 and R_7 resistors are found to be equal. Are R_4 and R_7 in series?

☐ Yes ☐ No

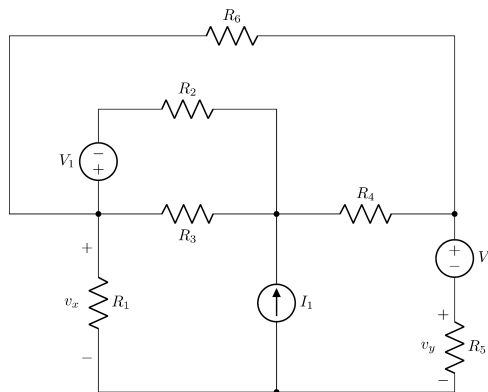
Justify your choice.

(b) R_1 , R_2 , and R_3 are connected in

☐ Series ☐ Parallel ☐ Neither series nor parallel ☐ Cannot be predicted

Explain your choice.

3.

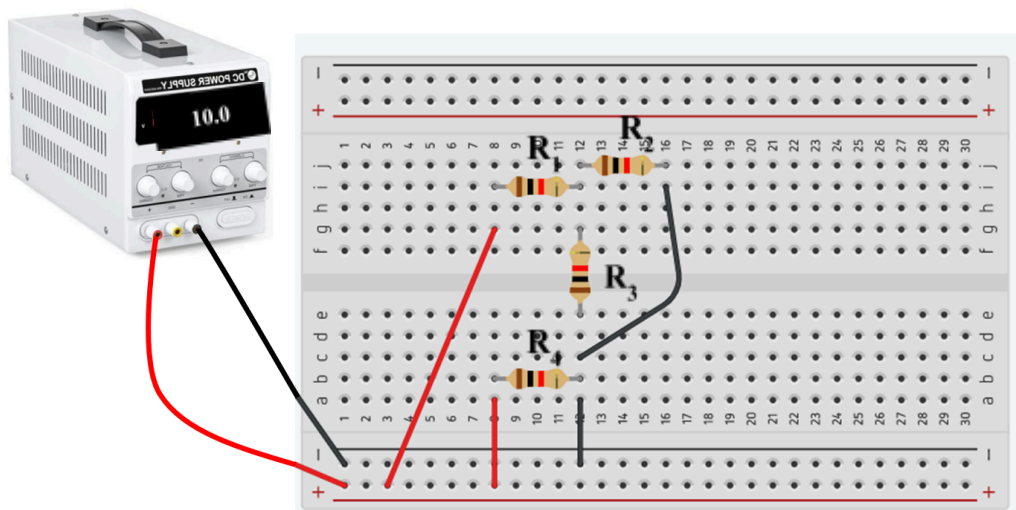


(a) If the voltages v_x and v_y are equal, are R_1 and R_5 in parallel?

☐ Yes ☐ No

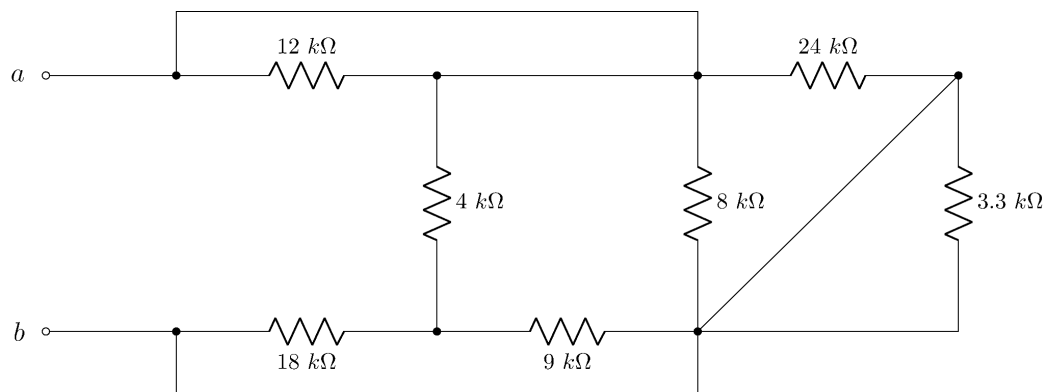
Justify your answer.

4.



Draw a circuit diagram of the circuit constructed on the breadboard above.

5. For the following circuit:

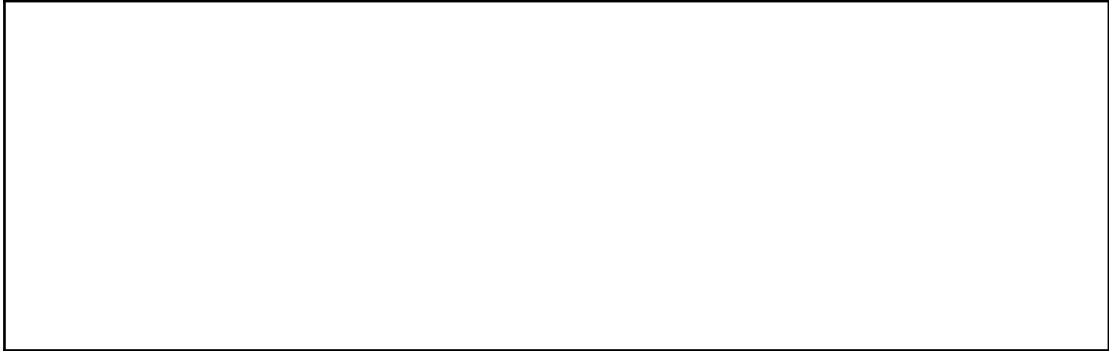


(a) How many nodes are there? Mark and label all the nodes in the circuit diagram.

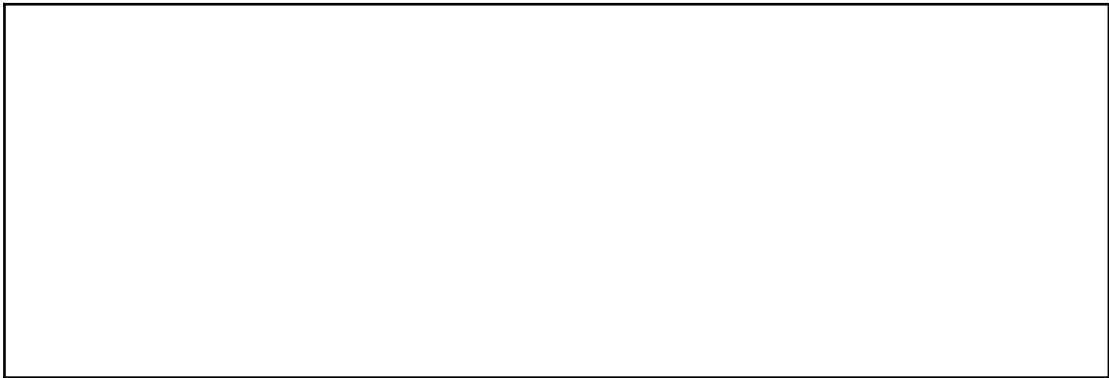
(b) Using the node labels in (a), fill out the table below by inputting the starting and ending nodes for each row that are connected to the corresponding circuit element.

Circuit Element	Starting/Ending Node	Ending/Starting Node
12 kΩ Resistor		
4 kΩ Resistor		
18 kΩ Resistor		
9 kΩ Resistor		
8 kΩ Resistor		
24 kΩ Resistor		
3.3 kΩ Resistor		

- (c) Based on the table in (b), draw a simplified version of the circuit using the labelled/identified nodes.

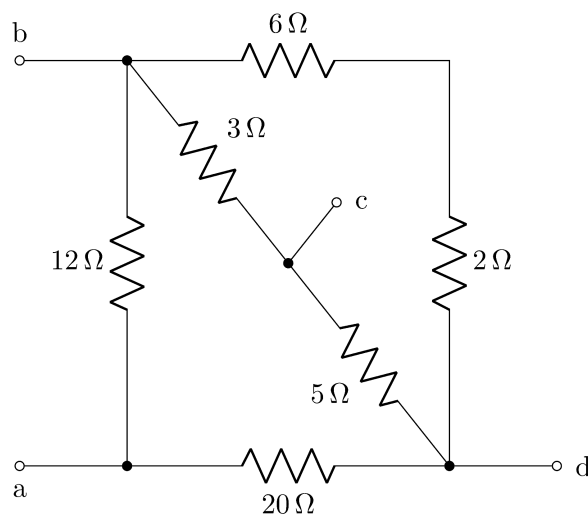


- (d) Determine the equivalent resistance between terminals a and b from the reduced circuit drawn in (c).



6. For the following circuit, determine R_{ab} , R_{ad} , R_{bd} and R_{bc} . Use logical operators to indicate the series-parallel combinations. For example, the following equation of R_{xy} means, two $10\ \Omega$ resistors are in parallel, their combination is in series with a $5\ \Omega$ resistor, and the total is again parallel with a $20\ \Omega$ resistor.

$$R_{xy} = \{(10 \parallel 10) + 5\} \parallel 20$$



$R_{ab} =$ $=$	$R_{ad} =$ $=$
$R_{bd} =$ $=$	$R_{bc} =$ $=$

Report

1. Fill in the theoretical parts of all the data tables.
2. Answers to the questions.
3. Discussion *[brief summary of the experiment, key concepts learned, accuracy and consistency of the measured data with theory, difficulties experienced and your thoughts on those, and ways to improve]*. Add pages if necessary.